RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I BASIC THEORY

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6011 Electromagnetic Interference and Compatibility
Name of the Faculty member: Mr.V.Rajesh
Name of the Topic: Case Histories
Name of the Innovative Practice: Videos
Date & Time: 09.08.2019 & 20 Minutes
ICT Tool Used:- Smart Class Room

Description of Case Histories Videos:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Goals (Learning Outcomes):

- The students will be able to analyze the history of EMI problems and root causes after playing video.
- The students will be able to identify the disadvantages of using EM radiation and unwanted emission of waves instead of using EM products in domestic and as well as industrial environments.

Use of appropriate methods:

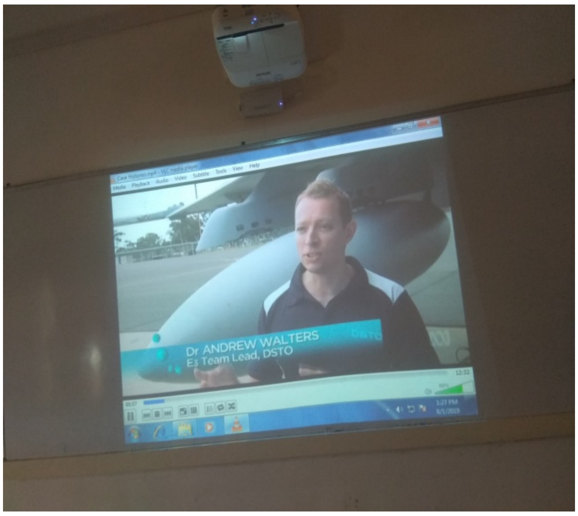
Justification for choosing Video:

Actually, students are not aware of Case histories. In real time, they have used more electronics and electrical products in their homes unknowingly. For that, I have chosen video method to create the awareness of the Case histories with the real time applications to the students.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution
Case Histories – Video



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ While showing the video, students identified the disadvantages of using EM radiation and unwanted emission of waves instead of using EM products in domestic and as well as industrial environments.
- ❖ The assessment of effectiveness of the activity was felt when asked Viva questions after showing the video.

<u>Reflective Critique:</u>

- **Benefits:**

The students are not aware of case histories of EM devices. During the theory class, some of the students are not interested to listen the class. So, I have shown video as histories of EM devices. After playing the video about case histories, students realized about the necessities of EM devices design and process.

- **Challenges:**

I have planned EM devices principles and operation nearly 10 minutes. But students are interested to see video and asked some doubts. It takes another 10 minutes for clarifying it.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1					2	1		

CO1: The student will be able to Find solution to EMI Sources, EMI problems in PCB level / Subsystem and system level design.

References:

1. Clayton Paul, "Introduction to Electromagnetic Compatibility", Wiley Interscience, 2006.
2. V Prasad Kodali, "Engineering Electromagnetic Compatibility", IEEE Press, Newyork, 2001.
3. https://www.youtube.com/watch?v=9Xgn40eGcqY

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

<u>Innovative Practices Description</u>

<u>UNIT II COUPLING MECHANISM</u>

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6011 Electromagnetic Interference and Compatibility
Name of the Faculty member: Mr.V.Rajesh
Name of the Topic: Automotive transients
Name of the Innovative Practice: Hardware Demonstration
Date & Time: 09.08.2019 & 20 Minutes
ICT Tool Used:- Smart Class Room

<u>Description of Hardware Demonstration:</u>

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

<u>Goals (Learning Outcomes):</u>

- ❖ The students will be able to analyze the fusing networks and isolators after demonstrating the Fuse networks and isolators.
- ❖ The students will be able to identify the advantages of using Miniature Circuit Breaker (MCB) instead of using fusing networks in domestic and as well as industrial environments.

<u>Use of appropriate methods:</u>

Justification for choosing Hardware Demonstration:

Actually, students are not aware of automotive transients. In real time, they have used automotive transients in their homes unknowingly. For that, I have chosen hardware demonstration method to create the awareness of the automotive transients with the real time applications to the students.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution
Automotive transients – Hardware Demonstration

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ While doing the demonstration, students identified the advantages of using Miniature Circuit Breaker (MCB) instead of using fusing networks in domestic and as well as industrial environments.
- ❖ The assessment of effectiveness of the activity was felt when asked Viva questions after demonstration

Reflective Critique:

- **Benefits:**

The students are not aware of automotive transient protection devices. During the theory class, some of the students are not interested to listen the class. So, I have shown MCB, Fuse and isolators as automotive transient devices. After demonstrated about Isolators and MCB's, students realized about the necessities of automotive transient protection devices.

- **Challenges:**

I have planned Miniature Circuit Breaker and Fuse demonstration nearly 10 minutes. But students are interested to see MCB, Fuse and isolators and asked some doubts. It takes another 10 minutes for clarifying it.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	2	1					2	1		

CO2: The student will be able to measure emission immunity level from different systems to couple with the prescribed EMC standards

References:

4. Clayton Paul, "Introduction to Electromagnetic Compatibility", Wiley Interscience, 2006.
5. V Prasad Kodali, "Engineering Electromagnetic Compatibility", IEEE Press, Newyork, 2001.
6. https://www.youtube.com/watch?v=9Xgn40eGcqY

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT III EMI MITIGATION TECHNIQUES

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6011 Electromagnetic Interference and Compatibility
Name of the Faculty member: Mr.V.Rajesh
Name of the Topic: Filter types and operation
Name of the Innovative Practice: MATLAB simulation
Date & Time: 28.08.2019 & 25 Minutes
ICT Tool Used:- Simulation Software

Description of MATLAB simulation:

One of the primary advantages of simulators is that they are able to provide users with practical feedback when designing real world systems. This allows the designer to determine the correctness and efficiency of a design before the system is actually constructed. Consequently, the user may explore the merits of alternative designs without actually physically building the systems. By investigating the effects of specific design decisions during the design phase rather than the construction phase, the overall cost of building the system diminishes significantly.

Goals (Learning Outcomes):

- The students will be able to design the different types of filters (Low pass, High pass, Band pass and Band elimination filter) in the MATLAB tool
- The students will be able to analyze the frequency response curves for the corresponding filters

Use of appropriate methods:

Justification for choosing MATLAB simulation:

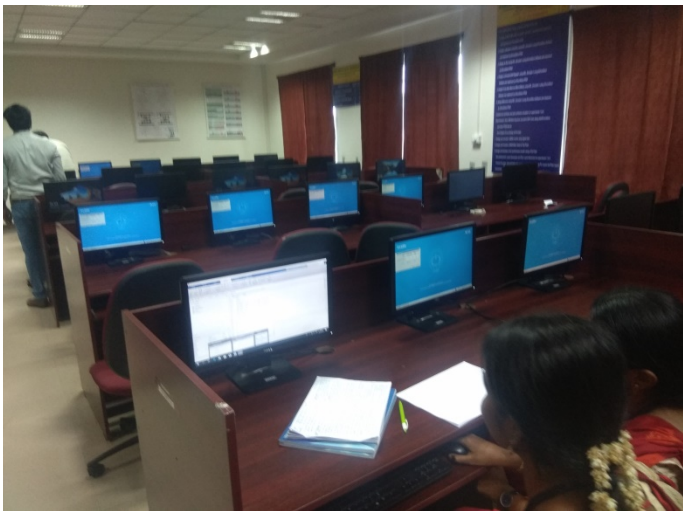
Already students have done MATLAB simulation in the Communication Systems laboratory and Digital Signal Processing laboratory. In DSP laboratory, they did filter related experiments. For that, I have chosen MATLAB simulation tool to refresh the design and frequency response analysis of filters which helps the students to understand the topic of Filter types and operation easily.

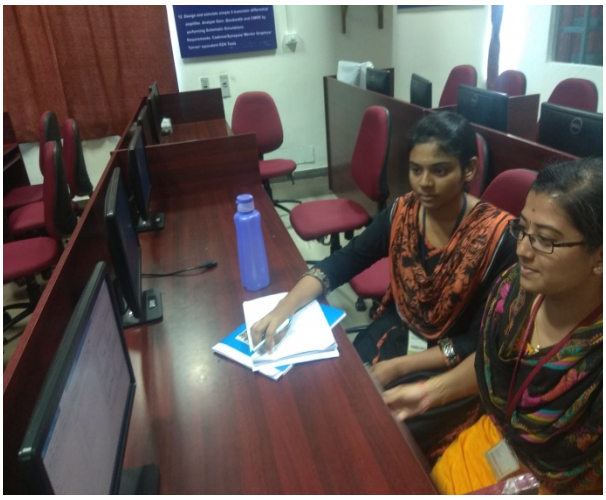
Effective presentation

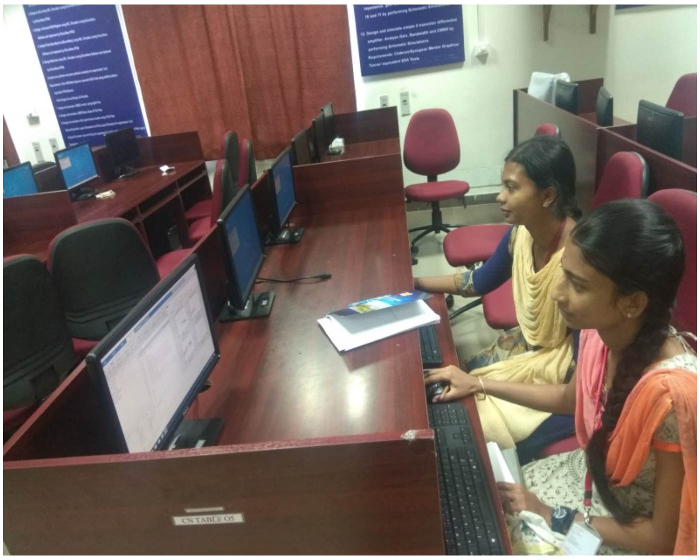
Details of the Implementation:

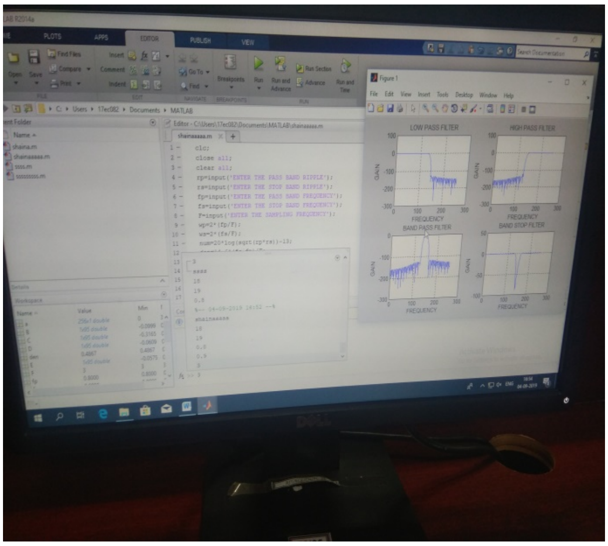
Innovative Teaching Method Execution

Filter types and operation – MATLAB simulation









Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- While doing the simulation, students analyzed the different types of filters (Low pass, High pass, Band pass and Band elimination filter) in the MATLAB tool
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly
- The assessment of effectiveness of the activity was felt when asked Viva questions after simulation

Reflective Critique:

- **Benefits:**

Some of the students struggled to precede the MATLAB simulation tool. I have explained the procedures once again. Some of them did syntax errors in the MATLAB tool. It takes more time to clear their errors own. After getting the frequency response of filters, students visualize the frequency response and they differentiate the different types of filters easily.

- **Challenges:**

I have planned MATLAB simulation tool for the topic of Filter types and operation nearly 15 minutes but I have taught MATLAB code for frequency response analysis of filters once again in the laboratory. I have taken 25 minutes to complete the MATLAB simulation.

Changes required for future (if required): NIL

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO3	2	2	2	1	3				2	1		

CO3: The student will be able to analyze the Mitigation techniques.

References:

7. Clayton Paul, "Introduction to Electromagnetic Compatibility", Wiley Interscience, 2006.
8. V Prasad Kodali, "Engineering Electromagnetic Compatibility", IEEE Press, Newyork, 2001.
9. https://in.mathworks.com/products/matlab.html

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT IV STANDARDS AND REGULATION

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6011 Electromagnetic Interference and Compatibility
Name of the Faculty member: Mr.V.Rajesh
Name of the Topic: Product Standards
Name of the Innovative Practice: One minute paper
Date & Time: 12.09.2019 & 5 Minutes
ICT Tool Used:- LCD Projector

Description of One minute paper Activity:

A "one-minute paper" may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day's lesson and provides the instructor with useful feedback.

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

Goals (Learning Outcomes):

- The students will be able to acquire knowledge about Product specific Standards.
- The students will be able to identify the European Standards (EN) for the respective subject areas of the products.

Use of appropriate methods:

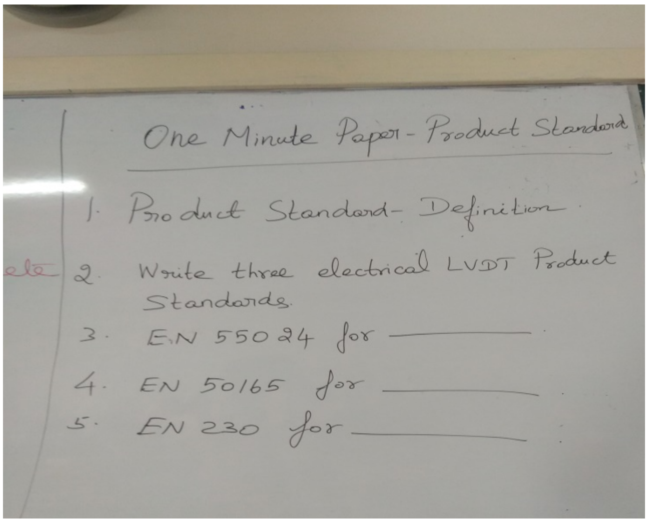
Justification for choosing One minute paper Activity:

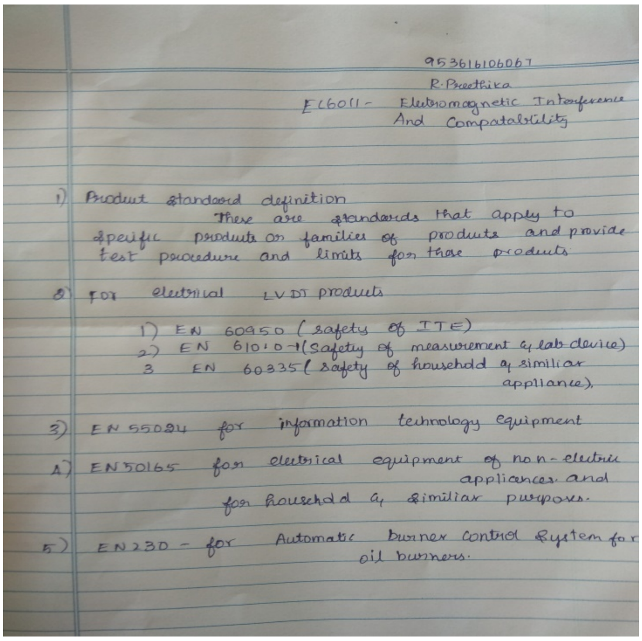
Students are not aware of the National and International Standards and regulating bodies. During the product standard class, more number of standards and their subject areas are given. For checking the listening skills of the students, I have chosen One minute paper activity for this topic.

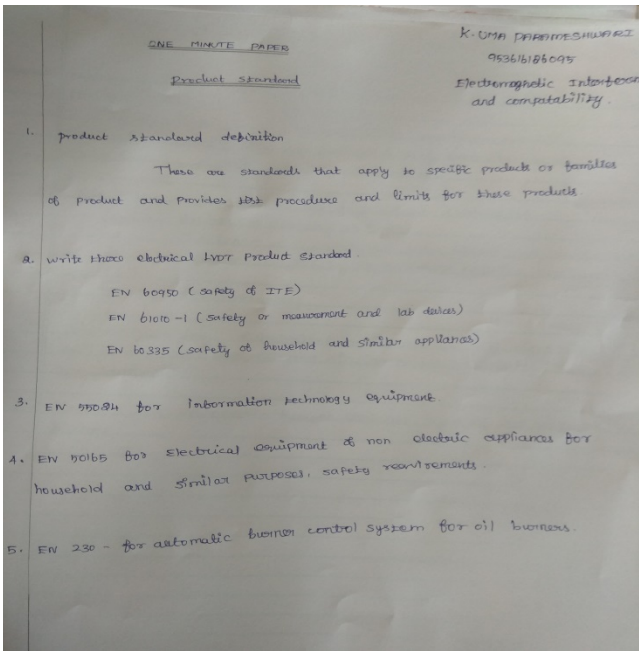
Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution
Product Standards – One minute paper







Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 70% of students answered correctly

- ❖ The assessment of effectiveness of the activity was felt when valued the one minute paper

Reflective Critique:

- **Benefits:**

Some of the student's performance was poor in the one minute paper activity but after completing the activity, they want to know the exact answers of product standards. Then we have discussed in the class room about product standards. After that, students felt the importance of listening the class.

- **Challenges:**

I have planned One minute paper activity nearly 2 minutes but it took 5 minutes because of giving the questions to the students and paper collection from the students.

Changes required for future (if required): NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2	1					2	1		

CO4: The student will be able to apply the data of EMI tests, and check the coupling and grounding tests.

References:

10. Clayton Paul, "Introduction to Electromagnetic Compatibility", Wiley Interscience, 2006.
11. V Prasad Kodali, "Engineering Electromagnetic Compatibility", IEEE Press, Newyork, 2001.
12. https://provost.tufts.edu/celt/files/MinutePaper.pdf

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

<u>Innovative Practices Description</u>

<u>UNIT V EMI TEST METHODS AND INSTRUMENTATION</u>

Degree, Semester& Branch: VII Semester B.E. ECE B
Course Code & Title: EC6011 Electromagnetic Interference and Compatibility
Name of the Faculty member: Mr.V.Rajesh
Name of the Topic: EMI test receivers, Spectrum analyzer
Name of the Innovative Practice: Videos
Date & Time: 28.09.2019 & 15 Minutes
ICT Tool Used:- LCD Projector

<u>Description of Videos:</u>

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

<u>Goals (Learning Outcomes):</u>

- ❖ The students will be able to acquire better knowledge about EMI test receivers and Spectrum analyzer
- ❖ The students will be able to identify the operation and working principle of EMI test receivers and Spectrum analyzer.

<u>Use of appropriate methods:</u>

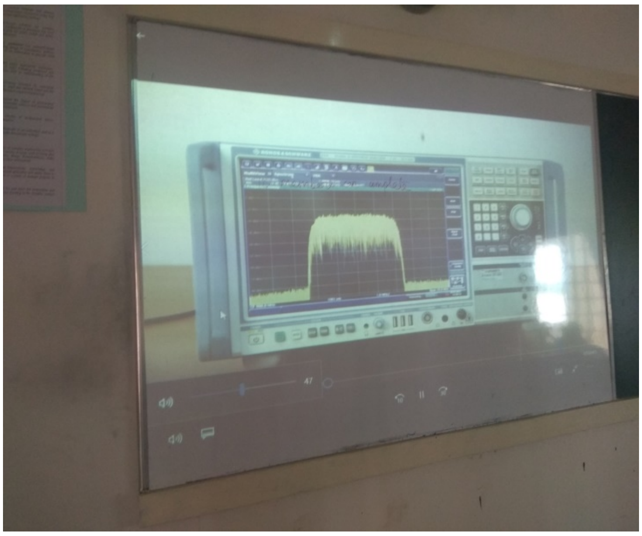
Justification for choosing Videos about EMI test receivers, Spectrum analyser:

Actually Spectrum analyser is not available in the laboratory for demonstrating. So that, I have downloaded the operation of spectrum analyser videos from the internet then I have shown it to the students.

Effective presentation

Details of the Implementation:

Innovative Teaching Method Execution
EMI test receivers, Spectrum analyzer – Videos



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 70% of students answered correctly
- ❖ The assessment of effectiveness of the activity was felt when asked Viva questions after shown the video

Reflective Critique:

- **Benefits:**

Some of the students did not understand the operation of Spectrum analyzer and they struggled to differentiate Spectrum analyzer with Cathode Ray Oscilloscope. After shown this video, they know the working principles of EMI test receivers and Spectrum analyzer effectively. Some of the students asked doubts about Spectrum analyzer and I clarified their doubts then there.

- **Challenges:**

I have shown 15 minutes video about the operation of EMI test receivers and Spectrum analyser to the students.

Changes required for future (if required): NIL

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO5	2	3	2	1					1	1		

CO5: The Students will be able to apply the various EMI test methods.

References:

13. Clayton Paul, “Introduction to Electromagnetic Compatibility”, Wiley Interscience, 2006.
14. V Prasad Kodali, “Engineering Electromagnetic Compatibility”, IEEE Press, Newyork, 2001.
15. https://www.youtube.com/watch?v=RqUQM2Onxlg
16. https://www.youtube.com/watch?v=nd0aM5jCnF4